Algorithm 2 Paths-to-graph

Input:
$\mathcal{P}$ A set of computational paths
Output: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ A computational graph formed from the paths of $\mathcal{P}$.
Initialize $\mathcal{S} \leftarrow \{\}$ {A set of already-seen computational path prefixes, to prevent us from double-counting attributions}
Initialize $\mathcal{V} \leftarrow \{\}$ {A dictionary mapping nodes to their attributions}
Initialize $\mathcal{E} \leftarrow \{\}$ {A dictionary mapping edges (node pairs) to their attributions}
for Each P in $\mathcal{P}$ **do**
  for $i \in [1 \dots |P|]$ **do**
    $s \leftarrow$ the prefix of P up to and including the i-th element
    if $s \in \mathcal{S}$ **then**
      Skip this iteration of the loop.
    end if
    Insert s into $\mathcal{S}$.
    if s has length 1 **then**
      Let n be the only node in s.
      Set $\mathcal{V}[n]$ to the attribution of n.
    else
      Set $n_{parent} \leftarrow P[i-1]$, $n_{child} \leftarrow P[i]$ {Earlier-layer nodes come later in the computational paths returned by Algorithm 1}
      Add the attribution of n_{child} to $\mathcal{V}[n_{child}]$
      Add the attribution of n_{child} to $\mathcal{E}[(n_{child}, n_{parent})]$
    end if
  end for
end for
return $\mathcal{V}, \mathcal{E}$

F Details on Section 4.1

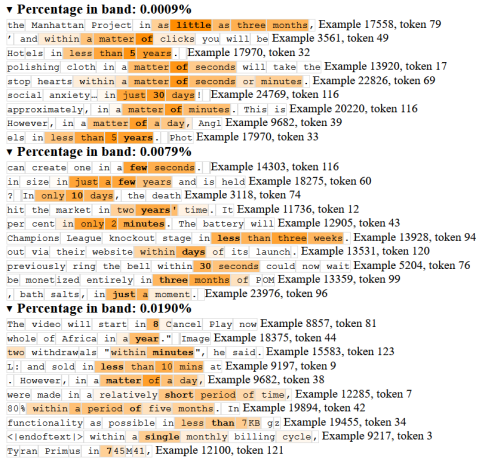
The transcoder used in the interpretability comparison was the Pythia-410M layer 15 transcoder trained with λ_1 sparsity coefficient 5.5×10^{-5} from Section 4.2. The SAE used in the comparison was a Pythia-410M layer 15 SAE trained on MLP inputs with $\lambda_1 = 7.0 \times 10^{-5}$. We used an SAE trained on MLP inputs rather than one trained on MLP outputs (as in § 4.2) because the interpretability comparison involves looking at which examples cause features to activate. This, in turn, is wholly determined by the encoder feature vectors. Because the transcoder's encoder feature vectors live in the MLP input space, it is thus most valid to compare the transcoder to an SAE whose encoder feature vectors also live in the MLP input space.

This transcoder-SAE pair was chosen because the transcoder and SAE sit at very similar points on the L_0-cross-entropy Pareto frontier: the transcoder has an L_0 of 44.04 and a cross-entropy of 3.35 nats, while the SAE has an L_0 of 47.85 and a cross-entropy of 3.36 nats. Pythia-410M was chosen as the model with the view that its features were likely to be more interesting than those of GPT2-small, while requiring less computational power to determine top activating examples than Pythia-1.4B would. Layer 15 was chosen largely heuristically, because we believed that this layer is late enough in the model to contain complex features, while not so late in the model that features are primarily encapsulating information about which tokens come next.

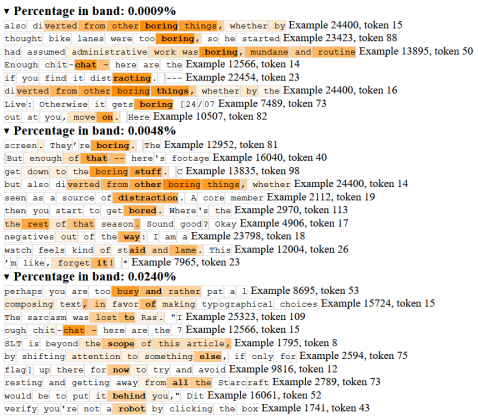
In Table 1, we refer to "context-free" features that interpretable features that seemed to fire on a single token (or two tokens) regardless of the context in which they appeared. Examples of features in all four categories ("interpretable", "maybe interpretable", "uninterpretable", and "context-free"), along with the exact annotation used by the human rater, can be found in Figure 6.

G Details on Section 5.2

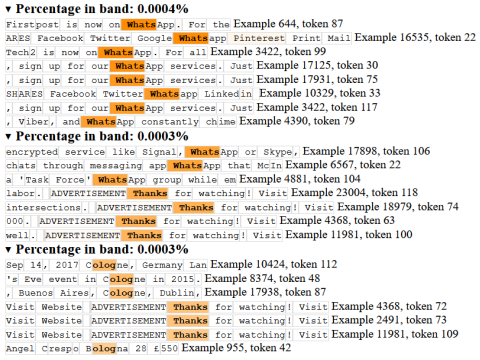
To obtain the de-embedding scores shown in Figures 5 and 4, the following method was used. First, we used the method presented in Appendix D.3 to determine which MLP0 transcoder features had the highest input-invariant connections to the given MLP10 transcoder feature through attention head 1 in



(a) Top-activating examples for a feature annotated as "interpretable". The specific annotation was `local context feature, fires on phrases describing short amounts of time`.



(b) Top-activating examples for a feature annotated as "maybe interpretable". The specific annotation was `local context feature for boredom? MAYBE`.



(c) Top-activating examples for a feature annotated as "uninterpretable". The specific annotation was `" Whats" > "ADVERTISEMENT Thanks" > "olog" NOT INTERPRETABLE`.

(d) Top-activating examples for a feature annotated as "context-free". The specific annotation was `"oc" in middle of words single-token feature`.

Figure 6: Examples of "feature-dashboards" used in the feature interpretation experiments.

layer 9. Specifically, for MLP0 transcoder feature i and MLP10 transcoder feature j, this attribution is given by $\left(\mathbf{f}_{\mathbf{dec}}^{(\mathbf{0},\mathbf{i})}\right)^T \left(\mathbf{W}_{\mathbf{OV}}^{(\mathbf{9},\mathbf{1})}\right)^T \mathbf{f}_{\mathbf{enc}}^{(\mathbf{10},\mathbf{j})}$. For each MLP10 transcoder feature, the top ten MLP0 transcoder features were considered. Then, for each MLP0 transcoder feature, the de-embedding score of each `YY` token for that MLP0 feature was computed. The total de-embedding score of each `YY` token for an MLP10 feature was computed as the sum of the de-embedding scores of that token over the top ten MLP0 features, with each de-embedding score weighted by the input-invariant attribution of the MLP0 feature. In Figures 5 and 4, the de-embedding scores were scaled and recentered in order to fit on the graph.

The *mean probability difference* metric discussed in the original greater-than work is as follows. Given the logits for each `YY` token, compute the softmax over these logits in order to obtain a probability distribution over the `YY` tokens; let p_y denote the probability of the token corresponding to year y. Then, the probability difference for a given prompt containing a certain input year y is given by

$\sum_{y'>y} p_{y'} - \sum_{y' \leq y} p_{y'}$. The mean probability difference is the mean of the probability differences over all 100 prompts.

H Full case studies

H.1 Classic blind case studies

H.1.1 Citation feature: `tc8[355]`

First, we checked activations for the first 12,800 prompts in the training data. Using this, we identified the prompt indexed at $(5701, 37)$ as one of 11 prompts for which `tc8[355]` activated above a score of 11.

Path-based analysis on input index $(5701, 37)$ revealed contributions from various tokens, notably `attn7[7]@35` and `attn5[6]@36`. However, we first decided to focus on the current token.

Current-token features. Top de-embeddings for both `tc0[9188]` and `tc0[16632]` were all variants of a semicolon: `;`, `';`, `%;`, and `.;`. We also checked `tc6[11831]@-1` and found that its top contributing features from layer 0 were `tc0[16632]` and `tc0[9188]`: the same two semicolon features. On the basis of this, we concluded that *the final token is a semicolon*.

Surname features. Next we focused on `attn7[7]@35`. Some interpretable features with high attributions through this component included `tc0[13196]@36` (years), `tc0[10109]@31` (open parentheses), `mlp8tc[355]attn7[7]attn0[1]@35` (components of last names), `tc0[12584]@32`: `P`, and `tc0[7659]@34`: `ck`.

Input-independent investigation of `tc6[21046]@35` revealed high contributions from `tc0[16382]` and `tc0[5468]`. feat016382 corresponded to tokens such as `oglu`, `owski`, and `zyk`; `tc0[5468]` corresponded to tokens such as `Burnett`, `Hawkins`, and `MacDonald`. Observing that all of these are (components of) surnames, we decided that *token 35 was likely (part of) a surname*.

Repeating analysis with prompt $(6063, 47)$**.** Top attributions for this prompt once again identified `tc0[9188]`, the semicolon feature from earlier. We filtered our computational paths to exclude this transcoder feature, since we already had a hypothesis about what it was doing. This identified `tc0[10109]@39` and `tc0[21019]@46` as top-contributing features.

The top de-embedding tokens for `tc0[10109]@39` were `(`, `(=`, and `(~`. On the basis of this, we determined that *token 39 was likely an open parenthesis*. Meanwhile, the top de-embedding tokens for `tc0[21019]@46` were `1983`, `1982`, and `1981`. This caused us to conclude that *token 46 was likely a year*.

We noted that, in the previous prompt, the attribution for the year features went through `attn5[6]`, whereas on this prompt it went through `attn2[9]`. We decided to investigate the behavior of `attn5[6]` on this prompt, and found that it was attributing to features `tc0[16542]@11`, `tc0[4205]@11`, and `tc0[19728]@11`. The de-embedding results for these were mixed: `tc0[16542]` were both close-parenthesis features, whereas `tc0[4205]` included citation-related tokens like `Accessed`, `Neuroscience`, and `Springer`.

Final result. We decided that `tc8[355]` was likely a semicolon-in-citations feature and looked at activating prompts. Top-activating prompts included "Res. 15, 241–247; 1978). In their paper, ", "aythamah , 2382; Tahdhīb al-", and "lesions (Poeck, 1969; Rinn, 1984). It". Note that the last of these was prompt $(5701, 37)$, i.e. the first case study we considered.

In general, the top-activating features corroborated our hypothesis, and we did not find any unrelated prompts. We noticed that many of the top activating prompts had a comma before the year in citations, but our circuit analysis never identified a comma feature.

We compared transcoder activations on the prompts "(Leisman, 1976;" and "(Leisman 1976;" and found `tc8[355]` to activate almost identically for both when all preceding MLPs were replaced by transcoders (4.855 and 4.906, respectively) and on the original model (12.484 and 12.13, respectively).